

CPU Scheduling Algorithms

Various CPU scheduling algorithms are-

1. **FCFS Scheduling**
2. **SJF Scheduling | SRTF Scheduling**
3. **Round Robin Scheduling**
4. **Priority Scheduling**

1. FCFS Scheduling-

- The process which arrives first in the ready queue is firstly assigned the CPU.
- In case of a tie, process with smaller process id is executed first.
- It is always non-preemptive in nature.

Advantages-

- It is simple and easy to understand.
- It can be easily implemented using queue data structure.
- It does not lead to starvation.

Disadvantages-

- It does not consider the priority or burst time of the processes.
- It suffers from **convoy effect**.

PRACTICE PROBLEMS BASED ON FCFS SCHEDULING-

Problem-01:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	4
P2	5	3
P3	0	2
P4	5	1
P5	4	3

If the CPU scheduling policy is FCFS, calculate the average waiting time and average turn around time.

Solution- Gantt Chart-



Gantt Chart

Here, black box represents the idle time of CPU.

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	7	$7 - 3 = 4$	$4 - 4 = 0$
P2	13	$13 - 5 = 8$	$8 - 3 = 5$
P3	2	$2 - 0 = 2$	$2 - 2 = 0$
P4	14	$14 - 5 = 9$	$9 - 1 = 8$
P5	10	$10 - 4 = 6$	$6 - 3 = 3$

Now,

- Average Turn Around time = $(4 + 8 + 2 + 9 + 6) / 5 = 29 / 5 = 5.8$ unit
- Average waiting time = $(0 + 5 + 0 + 8 + 3) / 5 = 16 / 5 = 3.2$ unit

Problem-02:

Consider the set of 3 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	2
P2	3	1
P3	5	6

If the CPU scheduling policy is FCFS, calculate the average waiting time and average turn around time.

Solution-

Gantt Chart-



Gantt Chart

Here, black box represents the idle time of CPU.

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	2	$2 - 0 = 2$	$2 - 2 = 0$
P2	4	$4 - 3 = 1$	$1 - 1 = 0$
P3	11	$11 - 5 = 6$	$6 - 6 = 0$

Now,

- Average Turn Around time = $(2 + 1 + 6) / 3 = 9 / 3 = 3$ unit
- Average waiting time = $(0 + 0 + 0) / 3 = 0 / 3 = 0$ unit

Problem-03:

Consider the set of 6 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	3
P2	1	2
P3	2	1
P4	3	4
P5	4	5
P6	5	2

If the CPU scheduling policy is FCFS and there is 1 unit of overhead in scheduling the processes, find the efficiency of the algorithm.

Solution-

Gantt Chart-



Gantt Chart

Here, δ denotes the context switching overhead.

Now,

- Useless time / Wasted time = $6 \times \delta = 6 \times 1 = 6$ unit
- Total time = 23 unit
- Useful time = 23 unit – 6 unit = 17 unit

Efficiency (η)

= Useful time / Total Total

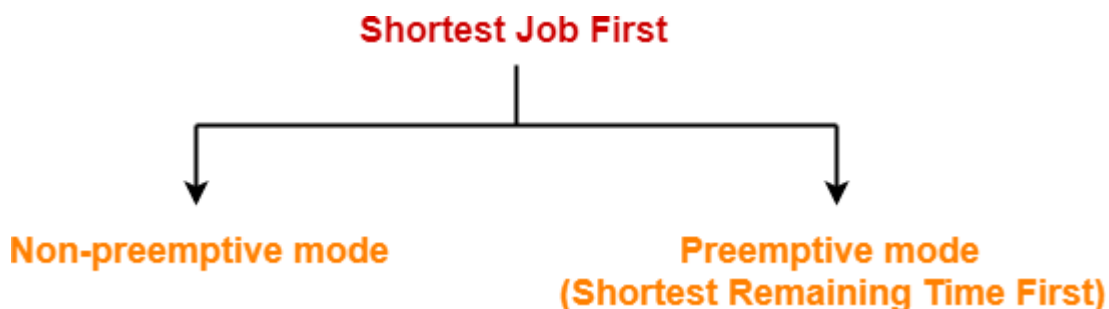
= 17 unit / 23 unit

= 0.739

2. SJF Scheduling-

In SJF Scheduling,

- Out of all the available processes, CPU is assigned to the process having smallest burst time.
- In case of a tie, it is broken by **FCFS Scheduling**.



- Scheduling can be used in both preemptive and non-preemptive mode.
- Preemptive mode of Shortest Job First is called as **Shortest Remaining Time First (SRTF)**.

Advantages-

- SRTF is optimal and guarantees the minimum average waiting time.
- It provides a standard for other algorithms since no other algorithm performs better than it.

Disadvantages-

- It can not be implemented practically since burst time of the processes can not be known in advance.
- It leads to starvation for processes with larger burst time.
- Priorities can not be set for the processes.
- Processes with larger burst time have poor response time.

PRACTICE PROBLEMS BASED ON SJF SCHEDULING-

Problem-01:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF non-preemptive, calculate the average waiting time and average turn around time.

Solution- Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	7	$7 - 3 = 4$	$4 - 1 = 3$
P2	16	$16 - 1 = 15$	$15 - 4 = 11$
P3	9	$9 - 4 = 5$	$5 - 2 = 3$
P4	6	$6 - 0 = 6$	$6 - 6 = 0$
P5	12	$12 - 2 = 10$	$10 - 3 = 7$

- Average Turn Around time = $(4 + 15 + 5 + 6 + 10) / 5 = 40 / 5 = 8$ unit
- Average waiting time = $(3 + 11 + 3 + 0 + 7) / 5 = 24 / 5 = 4.8$ unit

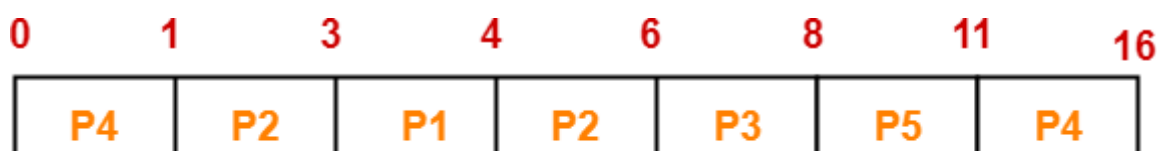
Problem-02:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF pre emptive, calculate the average waiting time and average turn around time.

Solution- Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	4	$4 - 3 = 1$	$1 - 1 = 0$
P2	6	$6 - 1 = 5$	$5 - 4 = 1$
P3	8	$8 - 4 = 4$	$4 - 2 = 2$
P4	16	$16 - 0 = 16$	$16 - 6 = 10$
P5	11	$11 - 2 = 9$	$9 - 3 = 6$

Now,

- Average Turn Around time = $(1 + 5 + 4 + 16 + 9) / 5 = 35 / 5 = 7$ unit
- Average waiting time = $(0 + 1 + 2 + 10 + 6) / 5 = 19 / 5 = 3.8$ unit

Problem-03:

Consider the set of 6 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	7
P2	1	5
P3	2	3
P4	3	1
P5	4	2
P6	5	1

If the CPU scheduling policy is shortest remaining time first, calculate the average waiting time and average turn around time.

Solution-

Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	19	$19 - 0 = 19$	$19 - 7 = 12$
P2	13	$13 - 1 = 12$	$12 - 5 = 7$
P3	6	$6 - 2 = 4$	$4 - 3 = 1$
P4	4	$4 - 3 = 1$	$1 - 1 = 0$
P5	9	$9 - 4 = 5$	$5 - 2 = 3$
P6	7	$7 - 5 = 2$	$2 - 1 = 1$

Now,

- Average Turn Around time = $(19 + 12 + 4 + 1 + 5 + 2) / 6 = 43 / 6 = 7.17$ unit
- Average waiting time = $(12 + 7 + 1 + 0 + 3 + 1) / 6 = 24 / 6 = 4$ unit

Problem-04:

Consider the set of 3 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	9
P2	1	4
P3	2	9

If the CPU scheduling policy is SRTF, calculate the average waiting time and average turn around time.

Solution-

Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	13	$13 - 0 = 13$	$13 - 9 = 4$
P2	5	$5 - 1 = 4$	$4 - 4 = 0$
P3	22	$22 - 2 = 20$	$20 - 9 = 11$

Now,

- Average Turn Around time = $(13 + 4 + 20) / 3 = 37 / 3 = 12.33$ unit
- Average waiting time = $(4 + 0 + 11) / 3 = 15 / 3 = 5$ unit

Problem-05:

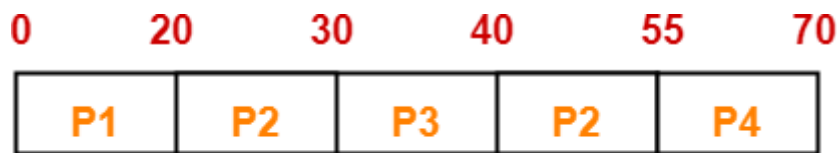
Consider the set of 4 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	20
P2	15	25
P3	30	10
P4	45	15

If the CPU scheduling policy is SRTF, calculate the waiting time of process P2.

Solution-

Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Thus,

- Turn Around Time of process P2 = 55 – 15 = 40 unit
- Waiting time of process P2 = 40 – 25 = 15 unit

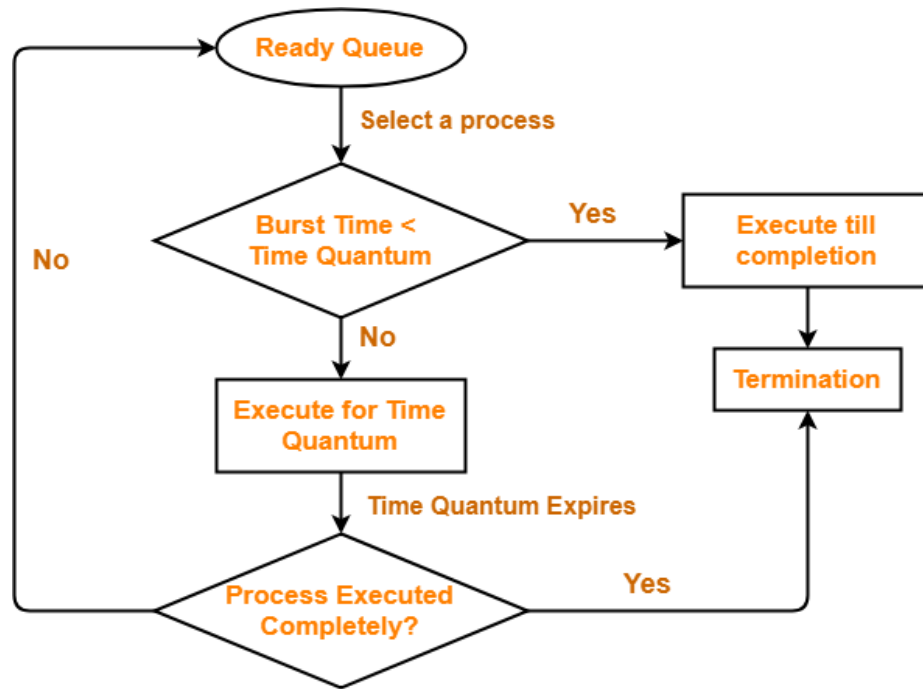
NOTE: Implementation of Algorithm-

- Practically, the algorithm can not be implemented but theoretically it can be implemented.
- Among all the available processes, the process with smallest burst time has to be selected.
- Min heap is a suitable data structure where root element contains the process with least burst time.

3. Round Robin Scheduling-

In Round Robin Scheduling,

- CPU is assigned to the process on the basis of FCFS for a fixed amount of time.
- This fixed amount of time is called as **time quantum** or **time slice**.
- After the time quantum expires, the running process is preempted and sent to the ready queue.
- Then, the processor is assigned to the next arrived process.
- It is always preemptive in nature.



Round Robin Scheduling

Advantages-

- It gives the best performance in terms of average response time.
- It is best suited for time sharing system, client server architecture and interactive system.

Disadvantages-

- Its performance heavily depends on time quantum.
- Priorities can not be set for the processes.

Important Notes-

Note-01:

With decreasing value of time quantum,

- Number of context switch increases
- Response time decreases
- Chances of starvation decreases

Note-02:

With increasing value of time quantum,

- Number of context switch decreases

- Response time increases
- Chances of starvation increases

Thus, higher value of time quantum is better in terms of number of context switch.

Note-03:

- With increasing value of time quantum, Round Robin Scheduling tends to become FCFS Scheduling.
- When time quantum tends to infinity, Round Robin Scheduling becomes FCFS Scheduling.

Note-04:

- The performance of Round Robin scheduling heavily depends on the value of time quantum.
- The value of time quantum should be such that it is neither too big nor too small.

PRACTICE PROBLEMS BASED ON ROUND ROBIN SCHEDULING-

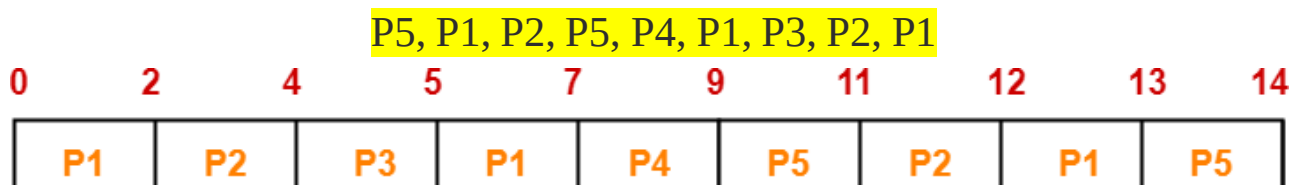
Problem-01:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

If the CPU scheduling policy is Round Robin with time quantum = 2 unit, calculate the average waiting time and average turn around time.

Solution-Ready Queue-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	13	$13 - 0 = 13$	$13 - 5 = 8$
P2	12	$12 - 1 = 11$	$11 - 3 = 8$
P3	5	$5 - 2 = 3$	$3 - 1 = 2$
P4	9	$9 - 3 = 6$	$6 - 2 = 4$
P5	14	$14 - 4 = 10$	$10 - 3 = 7$

Now,

- Average Turn Around time = $(13 + 11 + 3 + 6 + 10) / 5 = 43 / 5 = 8.6$ unit
- Average waiting time = $(8 + 8 + 2 + 4 + 7) / 5 = 29 / 5 = 5.8$ unit

Problem-02:

Consider the set of 6 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	4
P2	1	5
P3	2	2
P4	3	1
P5	4	6
P6	6	3

If the CPU scheduling policy is Round Robin with time quantum = 2, calculate the average waiting time and average turn around time.

Solution-

Ready Queue-

P5, P6, P2, P5, P6, P2, P5, P4, P1, P3, P2, P1



Gantt Chart

Process Id	Completion time	Turn Around time	Waiting time
P1	8	$8 - 0 = 8$	$8 - 4 = 4$
P2	18	$18 - 1 = 17$	$17 - 5 = 12$
P3	6	$6 - 2 = 4$	$4 - 2 = 2$
P4	9	$9 - 3 = 6$	$6 - 1 = 5$
P5	21	$21 - 4 = 17$	$17 - 6 = 11$
P6	19	$19 - 6 = 13$	$13 - 3 = 10$

Now,

- Average Turn Around time = $(8 + 17 + 4 + 6 + 17 + 13) / 6 = 65 / 6 = 10.84$ unit
- Average waiting time = $(4 + 12 + 2 + 5 + 11 + 10) / 6 = 44 / 6 = 7.33$ unit

Problem-03:

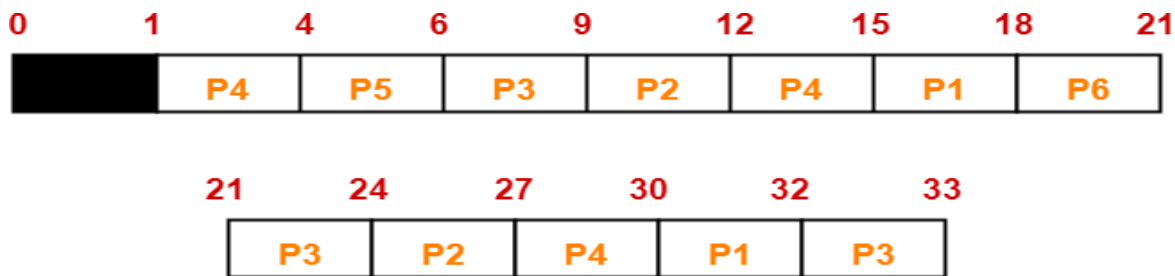
Consider the set of 6 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

If the CPU scheduling policy is Round Robin with time quantum = 3, calculate the average waiting time and average turn around time.

Solution-

Ready Queue- P3, P1, P4, P2, P3, P6, P1, P4, P2, P3, P5, P4



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	32	$32 - 5 = 27$	$27 - 5 = 22$
P2	27	$27 - 4 = 23$	$23 - 6 = 17$
P3	33	$33 - 3 = 30$	$30 - 7 = 23$
P4	30	$30 - 1 = 29$	$29 - 9 = 20$
P5	6	$6 - 2 = 4$	$4 - 2 = 2$
P6	21	$21 - 6 = 15$	$15 - 3 = 12$

Now,

- Average Turn Around time = $(27 + 23 + 30 + 29 + 4 + 15) / 6 = 128 / 6 = 21.33$ unit
- Average waiting time = $(22 + 17 + 23 + 20 + 2 + 12) / 6 = 96 / 6 = 16$ unit

Problem-04:

Four jobs to be executed on a single processor system arrive at time 0 in the order A, B, C, D. Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is-

1. 10
2. 4
3. 8
4. 9

Solution

Process Id	Arrival time	Burst time
A	0	4
B	0	1
C	0	8
D	0	1

Gantt chart-

Ready Queue-

C, A, C, A, C, A, D, C, B, A

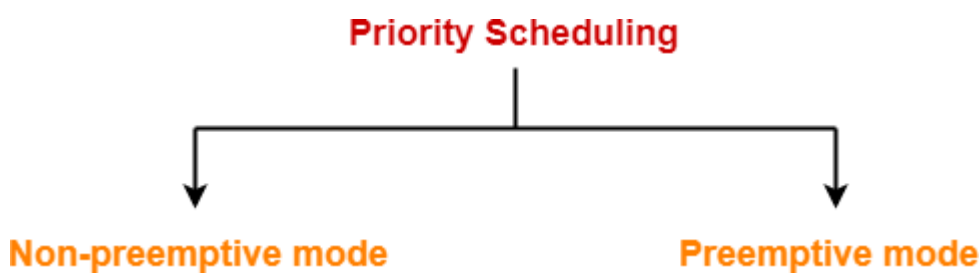


Gantt Chart

4. Priority Scheduling-

In Priority Scheduling,

- Out of all the available processes, CPU is assigned to the process having the highest priority.
- In case of a tie, it is broken by **FCFS Scheduling**.



- Priority Scheduling can be used in both preemptive and non-preemptive mode.

Advantages-

- It considers the priority of the processes and allows the important processes to run first.
- Priority scheduling in preemptive mode is best suited for real time operating system.

Disadvantages-

- Processes with lesser priority may starve for CPU.

Important Notes-

Note-01:

- The waiting time for the process having the highest priority will always be zero in preemptive mode.
- The waiting time for the process having the highest priority may not be zero in non-preemptive mode.

Note-02:

Priority scheduling in preemptive and non-preemptive mode behaves exactly same under following conditions-

- The arrival time of all the processes is same
- All the processes become available

PRACTICE PROBLEMS BASED ON PRIORITY SCHEDULING-

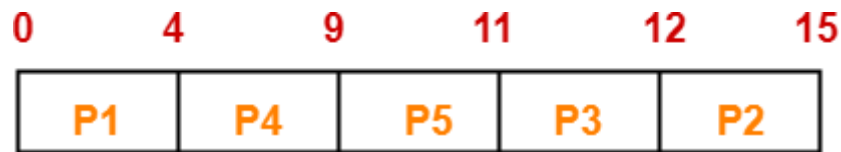
Problem-01:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

If the CPU scheduling policy is priority non-preemptive, calculate the average waiting time and average turn around time. (Higher number represents higher priority)

Solution- Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	4	$4 - 0 = 4$	$4 - 4 = 0$
P2	15	$15 - 1 = 14$	$14 - 3 = 11$
P3	12	$12 - 2 = 10$	$10 - 1 = 9$
P4	9	$9 - 3 = 6$	$6 - 5 = 1$
P5	11	$11 - 4 = 7$	$7 - 2 = 5$

Now,

- Average Turn Around time = $(4 + 14 + 10 + 6 + 7) / 5 = 41 / 5 = 8.2$ unit
- Average waiting time = $(0 + 11 + 9 + 1 + 5) / 5 = 26 / 5 = 5.2$ unit

Problem-02:

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

If the CPU scheduling policy is priority preemptive, calculate the average waiting time and average turn around time. (Higher number represents higher priority)

Solution-
Gantt Chart-



Gantt Chart

Now, we know-

- Turn Around time = Completion time – Arrival time
- Waiting time = Turn Around time – Burst time

Process Id	Completion time	Turn Around time	Waiting time
P1	15	$15 - 0 = 15$	$15 - 4 = 11$
P2	12	$12 - 1 = 11$	$11 - 3 = 8$
P3	3	$3 - 2 = 1$	$1 - 1 = 0$
P4	8	$8 - 3 = 5$	$5 - 5 = 0$
P5	10	$10 - 4 = 6$	$6 - 2 = 4$

Now,

- Average Turn Around time = $(15 + 11 + 1 + 5 + 6) / 5 = 38 / 5 = 7.6$ unit
- Average waiting time = $(11 + 8 + 0 + 0 + 4) / 5 = 23 / 5 = 4.6$ unit